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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations

May 2025

PH1007-1- PHYSICS OF MATERIALS

(For ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

List of constants:Velocity of light, $c = 3 \times 10^8 \text{ ms}^{-1}$, Planck's constant, $h = 6.63 \times 10^{-34} \text{ Js}$,Electron mass, $m = 9.11 \times 10^{-31} \text{ kg}$, Electron charge, $e = 1.6 \times 10^{-19} \text{ C}$,Permittivity of vacuum, $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$, Boltzmann constant, $k = 1.38 \times 10^{-23} \text{ J/K}$,Avogadro number, $N_A = 6.023 \times 10^{26} / \text{kg mole}$, Permeability of the free space, $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$.**PART - A: MULTIPLE CHOICE QUESTIONS****20 Marks**

1. **A system exhibiting SHM must possess**
A) Only inertia
B) elasticity, inertia and an external force
C) Only elasticity
D) Elasticity as well as inertia
2. **The restoring force in simple harmonic motion is directly proportional to**
A) Displacement
B) Velocity
C) Acceleration
D) Time period
3. **Any motion, which repeats itself at regular intervals according to a sinusoidal law, is called a**
A) Forced oscillation
B) Harmonic motion
C) Simple pendulum
D) Damped oscillation
4. **What is the frequency of SHM?**
A) Number of oscillations per unit time
B) Time for one oscillation
C) Time taken for motion to reverse direction
D) Same as angular frequency
5. **What is the effect of annealing on elasticity of materials?**
A) Increases the elasticity
B) Decreases the elasticity
C) Has no effect on elasticity
D) Distorts the material
6. **The ratio of lateral strain to linear strain is called**
A) Modulus of Elasticity
B) Modulus of Rigidity
C) Bulk Modulus
D) Poisson's Ratio
7. **Twisting moment is a product of _____ and the radius.**
A) Direction
B) Velocity
C) Force
D) Acceleration
8. **The Young's modulus of cantilever depends on the value of**
A) I_g
B) τ
C) L^3
D) All of these
9. **Fermi level for a metal is**
A) Highest energy level occupied by electrons at 0°C
B) Average value of all available energy levels
C) Highest energy level occupied by electrons at 0 K
D) Addition of energy of all available electron energy levels

10. As the doping to a pure semiconductor increases, the bulk resistance of the semiconductor
- A) Remains the same B) Increases
C) Decreases D) Becomes zero
11. Which of the following parameters can't be found with Hall Effect?
- A) Type of semiconductors (p or n type) B) Conductivity
C) Carrier concentration D) Area of the device
12. The relationship between current density J and electric field E is
- A) $J = \sigma E$ B) $J = \sigma / E$
C) $J = \sigma / 2E$ D) $J = 1 / \sigma E$
13. The Polarization that occur very rapidly
- A) Ionic B) Electronic
C) Orientation D) Space charge
14. The unit of dipole moment/unit volume is
- A) Coulomb/metre B) Coulomb/metre²
C) Coulomb/metre³ D) Coulomb
15. What changes in the capacitance of a capacitor occur if the dielectric material is removed?
- A) Increases B) Remains same
C) Decreases D) First increases and remains same.
16. Which of the following materials exhibit Ferro-electricity?
- A) Rochelle salt B) Platinum
C) Iron D) Hydrogen
17. A thermocouple must be able to provide
- A) small thermo e.m.f B) no thermo e.m.f
C) sufficient thermo e.m.f D) infinite thermo e.m.f
18. If the temperature of the cold junction of thermocouple is lowered, then the neutral temperature
- A) Increases B) Approaches inversion temperature
C) Decreases D) Remains the same
19. The net magnetic moment in the paramagnetic substance in the absence of the external magnetic field is
- A) Infinite B) Zero
C) One D) Negative value
20. The Magnetic materials characterized by high coercivity, high remanent magnetization and high hysteresis loss are known as
- A) Soft magnetic materials B) Hard magnetic materials
C) Demagnetized materials D) Both (A) and (B)

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) What is Simple Harmonic motion (SHM)? Derive the differential equation for SHM.	4	L*2	1	1,2
	b) Explain the theory of damped oscillation.	8	L2	1	1,2
	c) A spring is stretched 8 cm by a force of 10 N. Find the force constant. What will be the period of 4.0 kg mass suspended by it?	4	L3	1	1,2
2.	a) Explain the stress and strain diagram.	4	L1	2	1,2
	b) Define the terms Y (Young's Modulus), K (bulk modulus), n (rigidity modulus). Give the relation connecting these quantities.	8	L1, L2	2	1,2
	c) Calculate the extension produced in a wire of length 2 m and radius 0.013×10^{-2} m due to a force of 14.7 N applied along its length. Given Young's modulus of the material of the wire (Y) = 2.1×10^{11} N/m ² .	4	L3	2	1,2

3. a)	Explain the terms a) Poisson's ratio and b) Torsional pendulum.	4	L1	2	1,2
b)	Describe how the dimensions of a beam are altered when the beam is bent. Obtain an expression for the bending moment of the beam.	8	L2	2	1,2
c)	A body suspended symmetrically from the lower end of the wire of 1 m long, 1.22×10^{-3} m in diameter, oscillates about the wire as the axis with a period of 1.25 s. If the modulus of rigidity of the wire is $8 \times 10^{10} \text{ Nm}^{-2}$, calculate the moment of inertia of the body about the axis.	4	L3	2	1,2

Unit – II

4. a)	Explain Direct and Indirect bandgap semiconductors.	4	L1	3	1,2
b)	Explain Fermi factor. Discuss probability of occupancy of electrons at $T = 0\text{K}$, and $T > 0\text{K}$.	8	L2	3	1,2
c)	The compound gallium arsenide has an intrinsic conductivity of $10^{-6} \text{ ohm}^{-1} \text{ m}^{-1}$ at 20°C . How many electrons would have crossed the forbidden energy gap? [Given: $\mu_e = 0.88 \text{ m}^2\text{V}^{-1}\text{s}^{-1}$ and $\mu_h = 0.04 \text{ m}^2\text{V}^{-1}\text{s}^{-1}$].	4	L3	3	1,2
5. a)	Explain the effect of temperature on the conductivity of an intrinsic semiconductor.	4	L1	3	1,2
b)	What is Hall effect? Derive the expression for the Hall coefficient and Hall voltage and explain how they related to the mobility of the charge.	8	L1, L2	3	1,2
c)	The resistivity of germanium at 20°C is 0.5 ohmm . What will be its resistivity at 40°C if the band gap of germanium is 0.7 eV .	4	L3	3	1,2
6. a)	Explain the Frequency dependence of polarization with neat diagram.	4	L1	4	1,2
b)	Derive the expression for Clausius – Mossotti equation.	8	L2	4	1,2
c)	Calculate the polarization induced in the material due to application of electric field of 350 V/m . Given that the dielectric constant of the material is 4.2 .	4	L3	4	1,2

Unit – III

7. a)	Explain Seebeck effect and Peltier effect with neat diagram.	4	L1	5	1,2
b)	Derive an expression for thermo emf in terms of temperature.	8	L2	5	1,2
c)	The thermo emf of a thermocouple, one junction of which is kept at 0°C , is given by $e = 1600T - 4T^2$, where $T^\circ\text{C}$ is the temperature of hot junction. Calculate the neutral temperature.	4	L3	5	1,2
8. a)	Describe diamagnetic, paramagnetic and ferromagnetic materials.	4	L1	5	1,2
b)	Explain hysteresis loop (B – H Curve) with suitable diagram.	8	L2	5	1,2
c)	The magnetic field intensity in a piece of ferric oxide is 10^6 A/m . If the susceptibility of the material at room temperature is 1.5×10^{-3} , compute the flux density and magnetization of the materials.	4	L3	5	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
